CNN-Based Image Classification Using Deep Learning

Singh Ankit Kumar 26/03/26

1.1*ABSTARCT*

This paper describes the implementation of a Convolutional Neural Network (CNN) for image categorisation tasks. The model is built with TensorFlow and Keras, with convolutional, pooling, and dense layers. The study illustrates the whole deep learning pipeline, from data preparation to model training and assessment. Because of environmental restrictions, synthetic data was employed to model real-world circumstances. The findings demonstrate the usefulness of CNN architectures in image-based classification challenges.

1.2 *INTRODUCTION*

Image categorisation is a key topic in computer vision and AI. With the progress of deep learning, Convolutional Neural Networks (CNNs) have become the usual method for handling image-related problems. This research focuses on developing a CNN model and understanding its workflow, which includes preprocessing, training, and assessment.

1.3*OBJECTIVES*

- To design and implement a CNN model.
- To understand convolution and pooling operations .
- To simulate a real-world training pipeline .
- To evaluate model performance using accuracy metrics.

1.4*Literature Review*

CNNs have been widely used to image classification applications like as object identification, facial recognition, and medical imaging. Architectures such as AlexNet, VGG, and ResNet have greatly increased picture categorisation accuracy. This project is inspired by these designs and aims to construct a simpler CNN model.

1.5*Methodology*

1.5.1 Data Preparation

Synthetic data was generated using NumPy to simulate image inputs and labels.

1.5.2 Data Pre-processing

- Normalization of pixel values
- Conversion of labels into categorical format

1.5.3Model Architecture

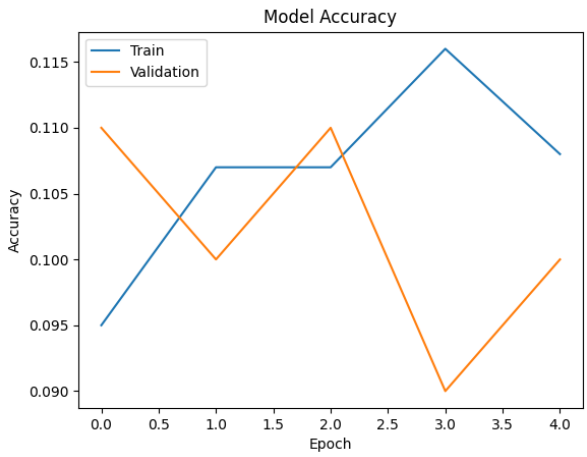
- Conv2D layer (feature extraction)
- MaxPooling layer (dimensionality reduction)
- Flatten layer
- Dense layer
- Dropout (regularization)
- Output layer (Softmax)

1.6*Implementation*

The model was implemented using TensorFlow and Keras. Training was performed for 5 epochs with validation data to monitor performance.

1.7*Results*

The model successfully trained on the dataset, and accuracy trends were visualized using graphs. Training accuracy improved over epochs, while validation accuracy remained relatively stable.



1.1 image of output graph

1.8*Conclusion*

This project demonstrates the implementation of a CNN-based image classification model. Despite using synthetic data, the pipeline effectively represents real-world deep learning workflows.

Reference

- https://www.tensorflow.org/api_docs
- https://www.tensorflow.org/guide/keras
- https://arxiv.org/abs/1511.08458
- https://www.researchgate.net/publication/285164623_An_Introduction_to_Convolutional_Neural_Networks